

1. Administrative & Study Identification

Full Study Title: An Open-label Study Evaluating Ofatumumab Treatment Effectiveness and PROs in Subjects With RMS Transitioning From Fumarate-based RMS Approved Therapies or Fingolimod to Ofatumumab **Official Title:** A Single-arm, Prospective, Multicentre, Open-label Study to Evaluate Ofatumumab Treatment Effectiveness and Patient-reported Outcomes (PRO) in Patients With Relapsing Multiple Sclerosis (RMS) Transitioning From Fumarate-based RMS Approved Therapies or Fingolimod **Short Title/Acronym:** ARTIOS Study **Protocol Number:** COMB157G23101 **NCT Number:** NCT04353492 **Sponsor/Company:** Novartis **Investigational Product:** Ofatumumab (Trade names: Kesimpta, Arzerra) **Phase of Development:** Phase 3 **Trial Status:** COMPLETED **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the treatment effectiveness of ofatumumab in reducing the Annualized Relapse Rate (ARR) in subjects with relapsing multiple sclerosis (RMS) over a 96-week period. **Secondary Objectives:** To assess the number of participants with Adverse Events (AEs) and Serious Adverse Events (SAEs), as well as magnetic resonance imaging (MRI) defined disease activity, confirmed disability worsening, and patient-reported outcomes (PROs).

2.2 Background & Rationale To address the need for highly effective disease-modifying therapies in patients with relapsing multiple sclerosis (RMS) who have experienced breakthrough disease (clinical relapses or new active MRI lesions) despite treatment with oral MS therapies, namely fingolimod or fumarate-based medications (e.g., dimethyl fumarate, diroximel fumarate, monomethyl fumarate). Transitioning to ofatumumab—a targeted B-cell depleting subcutaneous therapy—aims to provide robust control of inflammatory disease activity.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** 562 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 14-Jul-2020 **Estimated Primary Completion:** 09-Oct-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with a confirmed diagnosis of MS according to the 2017 Revised McDonald criteria, including relapsing forms of MS (RMS) and secondary progressive MS (SPMS). 2. Disability status at screening defined by an Expanded Disability Status Scale (EDSS) score of 0 to 4 (inclusive). 3. MS treatment history with a maximum of 3 Disease Modifying Therapies (DMTs), where all fumarates are considered as one DMT. 4. Currently transitioning from either fingolimod or a fumarate-based therapy (e.g., dimethyl fumarate [DMF], diroximel fumarate [DRF], monomethyl fumarate [MMF]) administered for a period of at least 6 months as their last DMT before first study drug administration. 5. Breakthrough disease activity while adequately using fumarates or fingolimod prior to transitioning for a minimum of 6 months, evidenced by $\geq$ 1 clinically reported relapse or $\geq$ 1 sign of Magnetic Resonance Imaging (MRI) activity (e.g., Gd⁺ enhancement, new or enlarging T2 lesions). 6. Neurologically stable within one month prior to the first study drug administration.

4.2 Exclusion Criteria 1. Subjects with primary progressive MS, SPMS without disease activity, or meeting criteria for neuromyelitis optica. 2. Disease duration of more than 10 years since MS diagnosis. 3. Pregnant or nursing (lactating) women, and women of child-bearing potential unless they are using highly effective forms of contraception during dosing and for at least 6 months after stopping study medication. 4. Subjects with active chronic disease of the immune system other than MS or with immunodeficiency syndrome (including Acquired Immunodeficiency Syndrome [AIDS]). 5. Active systemic bacterial, fungal, or viral infections (such as hepatitis B and C, HIV, COVID-19), or neurological symptoms consistent with Progressive Multifocal Leukoencephalopathy (PML) or with confirmed PML. 6. Subjects at risk of developing or having reactivation of syphilis or tuberculosis (e.g., subjects with known exposure to, or history of syphilis, or active/latent tuberculosis, even if previously treated). 7. Received any live or live-attenuated vaccines within 4 weeks prior to first study drug administration, or have been treated with prohibited medications within specified timeframes (e.g., corticosteroids, rituximab, ocrelizumab, alemtuzumab, natalizumab, daclizumab, cyclophosphamide, teriflunomide). 8. Subjects suspected of not being able or willing to cooperate or comply with study protocol requirements in the opinion of the investigator.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ofatumumab administered once every 4 weeks. **Route of Administration:** Subcutaneous injection (via a pre-filled pen). **Duration of Treatment:** 96 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard symptomatic therapies for the management of MS-related symptoms. **Prohibited:** Concurrent use of other disease-modifying MS therapies (e.g., rituximab, ocrelizumab, alemtuzumab) and receipt of any live or live-attenuated vaccines within 4 weeks prior to the first dose of the study drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, MRI, EDSS, Safety Labs
Baseline	Day 0	Transition off prior therapy, First Dose of Ofatumumab, PRO questionnaires
Interim	Every 4 Weeks	Subcutaneous injection, Safety Labs, Relapse Assessment, Ongoing PRO tracking
End of Study	Week 96	Final EDSS evaluation, Exit MRI (T1/T2 lesions), Safety wrap-up, Treatment effectiveness analysis

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized Relapse Rate (ARR) across the 96-week study period. **Measurement Method:** Calculated based on the cumulative number of confirmed relapses by patient, adjusted for time-in-study. Confirmed relapses are those accompanied by a clinically relevant change in the Expanded Disability Status Scale (EDSS). **Analysis:** The primary analysis describes the ARR with a one-sided 95% confidence bound and tests the null hypothesis (H_0): $ARR \geq 0.18$ versus the alternative hypothesis (H_1): $ARR < 0.18$.

7.2 Secondary & Exploratory Endpoints

- Number of Participants With Adverse Events (AEs) and Serious Adverse Events (SAEs).
- Magnetic Resonance Imaging (MRI) defined disease activity, including the number of T1 Gd-enhancing lesions per scan and annualized rate of new or enlarging T2 lesions.
- 3-month and 6-month Confirmed Disability Worsening (CDW3 and CDW6).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during study visits focusing specifically on secondary outcome safety endpoints (number of AEs/SAEs), injection-related systemic reactions, infections, laboratory test abnormalities, and vital signs. **Serious Adverse Events (SAEs):** Routine SAE reporting protocols implemented for immediate tracking of severe events. **Data Monitoring Committee (DMC):** Independent medical safety review board overseeing patient tolerability and compliance.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary effectiveness outcomes (ARR). Safety Set for all adverse event incidence reporting. **Sample Size Justification:** Target enrollment of 562 subjects to ensure adequate statistical power to evaluate the reduction of relapse rates compared to baseline pre-transition. **Handling of Missing Data:** The ARR was estimated by fitting a negative binomial regression model with a log-link, adjusting for factors such as prior MS therapies, previous year relapses, baseline EDSS, baseline T1 Gd-enhancing lesions, and age at baseline.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote clinical monitoring to guarantee protocol adherence and PRO data validity. **Audit Trail:** Validated Electronic Data Capture (EDC) systems utilized for secure eCRF locking and transparent data clarification processing.

11. Study Identifiers & Governance

Primary ID: COMB157G23101 **Registry Number:** NCT04353492 **Sponsor/Lead Organization:** Novartis **Study Title:** ARTIOS Study **Official Regulatory Title:** A Single-arm, Prospective, Multicentre, Open-label Study to Evaluate Ofatumumab Treatment Effectiveness and Patient-reported Outcomes (PRO) in Patients With Relapsing Multiple Sclerosis (RMS) Transitioning From Fumarate-based RMS Approved Therapies or Fingolimod

12. Clinical Framework (RMS Specific)

Primary Condition: Relapsing Multiple Sclerosis (RMS) **Diagnosis Threshold:** Clinically confirmed diagnosis of MS (2017 Revised McDonald criteria) with breakthrough disease on previous oral therapy (fingolimod or fumarates). **Medical Methodology:** Targeted B-cell depletion therapy. **Optimization Target:** Complete suppression of clinical relapses and MRI lesion activity.

13. Study Procedures & Methodology

Management Pathway: Step-down/transition protocol from prior oral DMTs to subcutaneous ofatumumab. **Education Protocol (HCP):** Site-level injection technique training, PML screening, and systemic infection management guidelines. **Education Protocol (Patient):**

Self-injection training utilizing the ofatumumab Sensoready® pen and PRO survey completion instructions. **Quality Audit Mechanism:** Comprehensive remote tracking of patient adherence, injection success, and PRO survey completion rates.

14. Observation & Follow-Up Schedule

Total Duration: 96 weeks **Onsite Visit Cadence:** Pre-defined schedule aligning with screening, baseline, and significant MRI/EDSS milestone evaluations (e.g., Weeks 24, 48, 96). **Remote Monitoring Frequency:** Continuous evaluation throughout the 4-week self-administration intervals. **Checklist Review Interval:** Regular EDSS assessment and symptom checklist reviews scheduled with clinical follow-ups.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				